

Tulasi Venkata Sri Varshini Padamata

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EDUCATION

Rochester Institute of Technology

Masters in Artificial Intelligence

Specialization: Machine Learning Applications, Distributed Systems, Engineering Scalable Systems

August 2024 - May 2026

Rochester, NY

RMK Engineering College

Bachelors in computer science and engineering

August 2019 - May 2023

India

SKILLS AND CERTIFICATIONS

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision, NLP/NLU, Generative AI, Prompt Engineering, Deep Learning

Cloud & MLOps: AWS (Bedrock, SageMaker, EKS, Lambda, CloudWatch, DynamoDB), Docker, Kubernetes, MLflow, Terraform, CI/CD Pipelines, Model Context Protocol (MCP), Palantir Foundry, Snowflake

Programming & Data: Python, PySpark, SQL, LangChain, FastAPI, Pandas, NumPy, PostgreSQL, MongoDB, Git, GitHub Actions

EXPERIENCE

NOVACIS DIGITAL

Machine Learning Engineer

India
November 2023 – May 2024

- Built a SageMaker-based document AI system with LayoutLM + BERT, with 18% F1 improvement on 10K+ legal documents
- Built and deployed an OCR + LLM pipeline on Amazon EKS, shrinking review time to less than 10 minutes per document
- Built a FAISS-based semantic search system, delivering sub-second (300 ms) retrieval and cutting analyst lookup time by 60%
- Benchmarked GPT-4 vs Claude for contract analysis, reducing inference cost by 35% while maintaining response reliability
- Collaborated cross-functionally to productionize AWS ML pipelines, enabling reliable batch and real-time workflows

RMK Research Lab

Machine Learning Intern

India
August 2023 – January 2024

- Developed an explainable ECG-based ML pipeline with ensemble models and LIME, backed by a centralized feature store
- Optimized ensemble classifiers (XGBoost, RF, GB), delivering 6–8% accuracy gains over individual models
- Built reproducible ML pipelines for preprocessing, training, and validation on 1,190+ clinical records
- Achieved 93.1% accuracy and 0.95 ROC-AUC, outperforming AdaBoost and CatBoost while maintaining model stability

PUBLICATIONS

Lightweight Channel Attention for Efficient CNNs - ([Paper Link](#))

- Designed and evaluated a lightweight channel attention module (LCA) achieving competitive accuracy with negligible parameter and latency overhead on ResNet-18 and MobileNetV2

EDAT-Lite: Predictive Difficulty-Aware Training for Neural Networks - ([Paper Link](#))

- Proposed a lightweight difficulty-aware training framework that learns and predicts hard examples using an auxiliary head, improving interpretability of training dynamics while preserving accuracy on MNIST

PROJECTS

Real-Time Video Search Platform - ([GitHub](#))

- Engineered CLIP-based video search indexing 25 videos with 380+ scene embeddings; achieved 1.2s query latency
- Built automatic scene detection pipeline at 0.4x speed; achieved 78% boundary accuracy vs manual annotation on validation set
- Deployed full-stack application with React frontend; delivered timestamped results with 0.82 similarity score for top-ranked scenes
- Created a search interface supporting 15+ queries; demonstrated multimodal retrieval combining visual and text embeddings

Real-Time Feature Store & ML Serving Platform - ([GitHub](#))

- Architected streaming feature pipeline with Feast and Kafka processing 5,000 events/hour at 45ms latency
- Deployed 3 models (fraud: 88%, recommendation: 0.76 NDCG@5, forecasting: 12% MAPE) sharing 8 features via unified store
- Built monitoring with MLflow, Kubernetes metrics, triggering auto-retraining on performance degradation
- Implemented feature drift detection and data quality checks; automated alerts for distribution shifts across 8 production features

Multi-Agent Document Intelligence System - ([GitHub](#))

- Built multi-agent system automating text extraction; achieved 82% accuracy on 150-document test set with processing time < 5s
- Implemented RAG with Qdrant processing 500+ chunks; enabled semantic search with 85% top-3 relevance across 50 queries
- Developed production FastAPI service handling 3 document types; reduced manual entry time 84% reduction
- Deployed the model with monitoring and validation pipelines to ensure stable production performance

CERTIFICATIONS

- Nvidia Edge AI Specialization
- Stanford Machine Learning Specialization